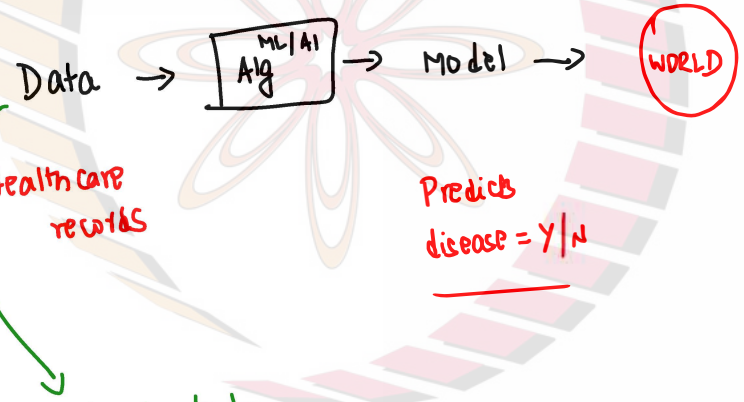
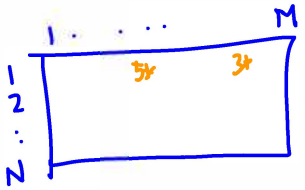
Note Title

PRIVACY IN AI/ML



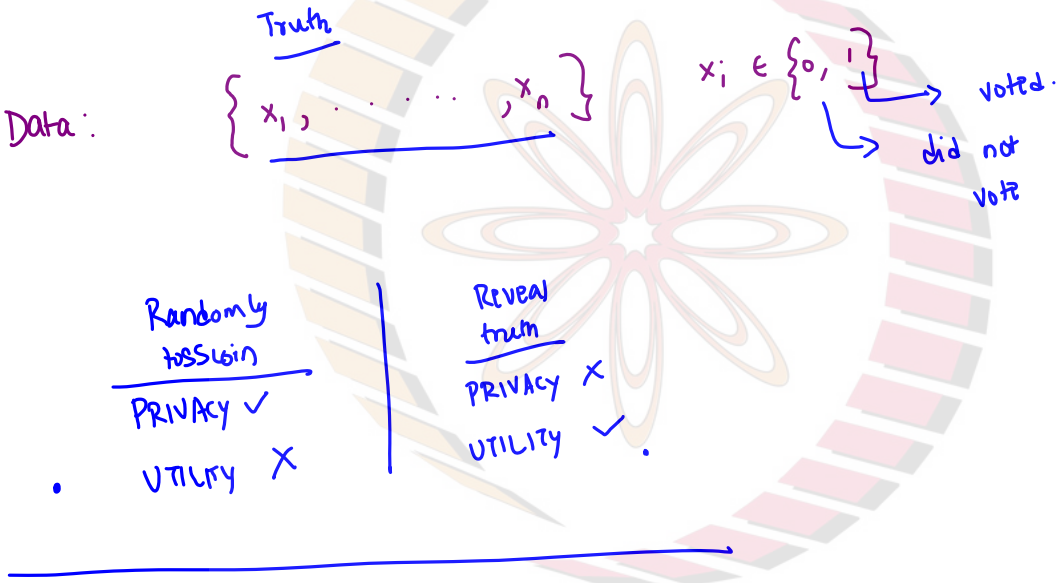
Anonymisation

Netflix



- CRYPTOGRAPHIC SOLUTION ✗
- STATISTICAL SOLUTION TO PRIVACY ✓

Note Title



RANDOMIZED RESPONSE

Note Title

Truth $\{x_1, \cdots, x_n\}$ $x_i \in \{0,1\}$

Revealed $y_i = \begin{cases} x_i & \text{with probability } \frac{e^{\epsilon}}{1+e^{\epsilon}} \quad \text{TRUTH} \\ 1-x_i & \text{with probability } \frac{1}{1+e^{\epsilon}} \quad \text{FALSEHOOD} \end{cases}$

Consider two databases

$X = \{x_1, \cdots, x_n\}$

$X' = \{x_1', \cdots, x_n'\}$

Assume $x_i = x_i' \quad \forall i \in \{1, \cdots, n-1\}$

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Note Title

Revealed

$RR(x) = y = [y_1, \cdots, y_n]$

$RR(x') = y' = [y_1', \cdots, y_n']$

$y_i \in \{0,1\}$

$y_i' \in \{0,1\}$

$$Prob\left(RR(x) = b\right) = Prob\left([y_1, y_2 \cdots y_n] = [b_1, b_2 \cdots, b_n]\right)$$

$[b_1\ b_2 \cdots b_n]$

eg: $[0\ 1\ 0\ 1\ 1]$

$$= \prod_{i=1}^{n} P(y_i = b_i) \quad \left[\text{Independence of } y_1, \cdots, y_n\right]$$

$$= \left[\prod_{i=1}^{n-1} P(y_i = b_i)\right] P(y_n = b_n) \quad -(1)$$

Note Title

$$\text{Prob}\left(RR(x') = b\right) = \prod_{i=1}^{n} P\left(y_i' = b_i\right)$$

$$= \left(\prod_{i=1}^{n-1} P\left(y_i' = b_i\right)\right) P\left(y_n' = b_n\right)$$

$$= \left(\prod_{i=1}^{n} P\left(y_i = b_i\right)\right) P\left(y_n' = b_n\right) \quad \text{— (2)}$$

$$\frac{P\left(y_n = b_n\right)}{P\left(y_n' = b_n\right)} = \begin{cases} \dfrac{e^{\epsilon}/(1+e^{\epsilon})}{1/(1+e^{\epsilon})} = e^{\epsilon} & b_n = x_n \\ \dfrac{1/(1+e^{\epsilon})}{1/(1+e^{\epsilon})/e^{\epsilon}/(1+e^{\epsilon})} = e^{-\epsilon} & b_n \neq x_n \ (\text{i.e., } b_n = 1 - x_n) \end{cases}$$

Note Title

In both cases,

$$\frac{P(y_n = b_n)}{P(y_n' = b_n)} \leq e^{\epsilon}$$

$$\Rightarrow \boxed{P(y_n = b_n) \leq e^{\epsilon} \cdot P(y_n' = b_n)} \quad \text{— (3)}$$

$$\frac{Pr(RR(x) = b)}{Pr(RR(x') = b)} = \frac{P(y_n = b_n)}{P(y_n' = b_n)} \leq e^{\epsilon}$$

(1) & (2) (3)

Note Title

$$Pr(RR(x)=b) \leq e^{\epsilon} Pr(RR(x')=b)$$

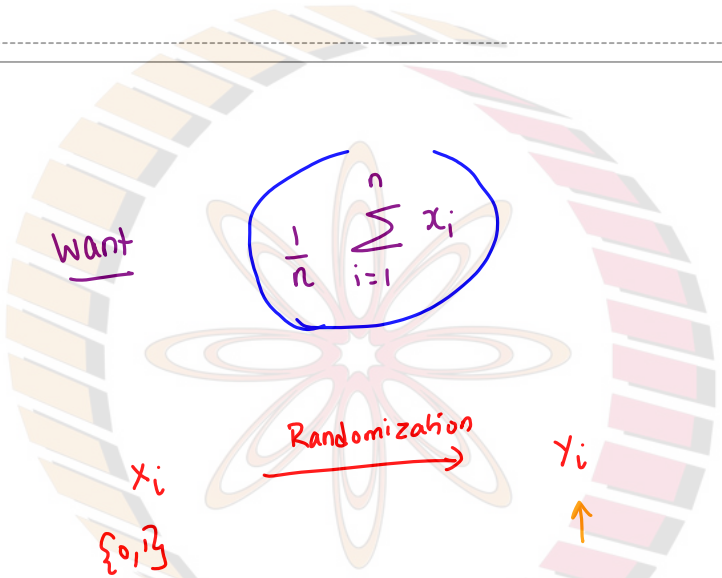
observe

$$\Rightarrow \boxed{Pr(RR(x')=b) \cdot e^{-\epsilon} \leq Pr(RR(x)=b) \leq e^{\epsilon} Pr(RR(x')=b)}$$

UTILITY of the mechanism

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Note Title



$$\mathbb{E}[y_i] = \left(\frac{e^{\epsilon}}{1+e^{\epsilon}}\right) x_i + \left(\frac{1}{1+e^{\epsilon}}\right)(1-x_i)$$

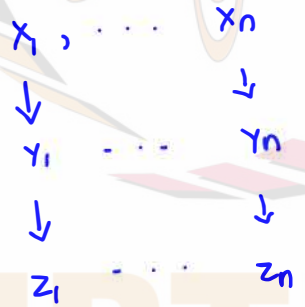
$$\frac{e^{\epsilon} x_i + 1 - x_i}{1+e^{\epsilon}} = x_i\left(\frac{e^{\epsilon}-1}{e^{\epsilon}+1}\right) + \frac{1}{1+e^{\epsilon}}$$

Note Title



$$z_i = \left(y_i - \frac{1}{1+e^{\epsilon}}\right) \cdot \left(\frac{e^{\epsilon}+1}{e^{\epsilon}-1}\right)$$

$\mathbb{E}[z_i] = x_i$ [Exercise]



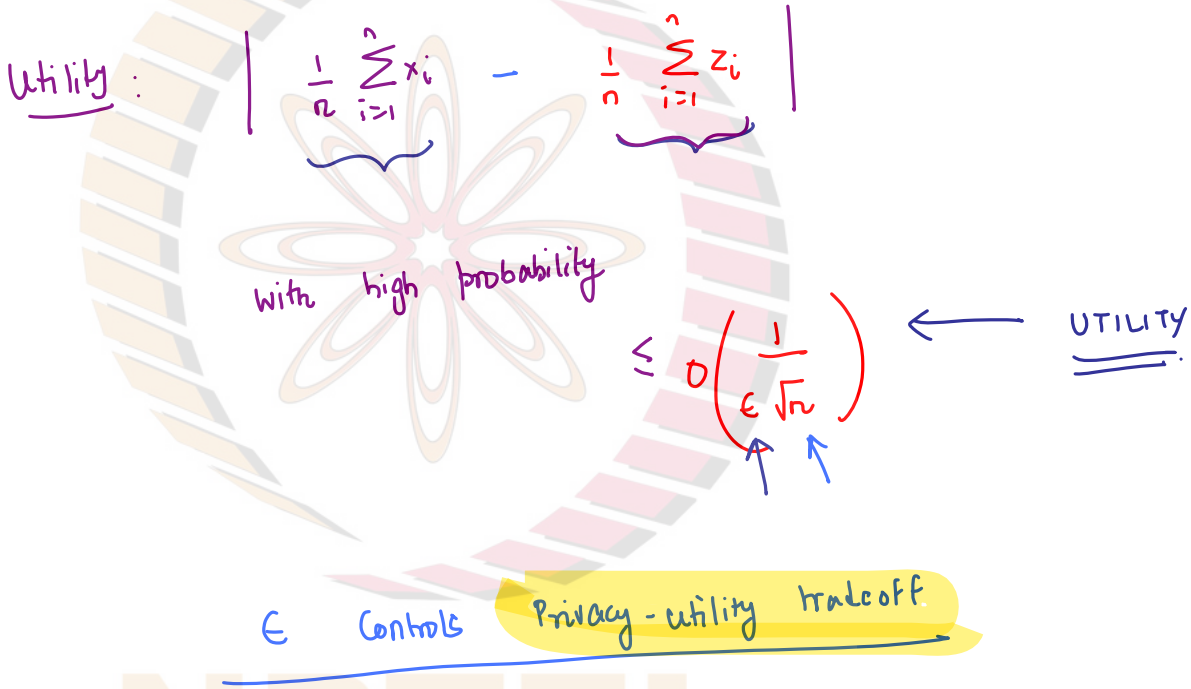
$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

Guess for $\bar{x}$ = $\bar{z} = \frac{1}{n}\sum_{i=1}^{n} z_i$

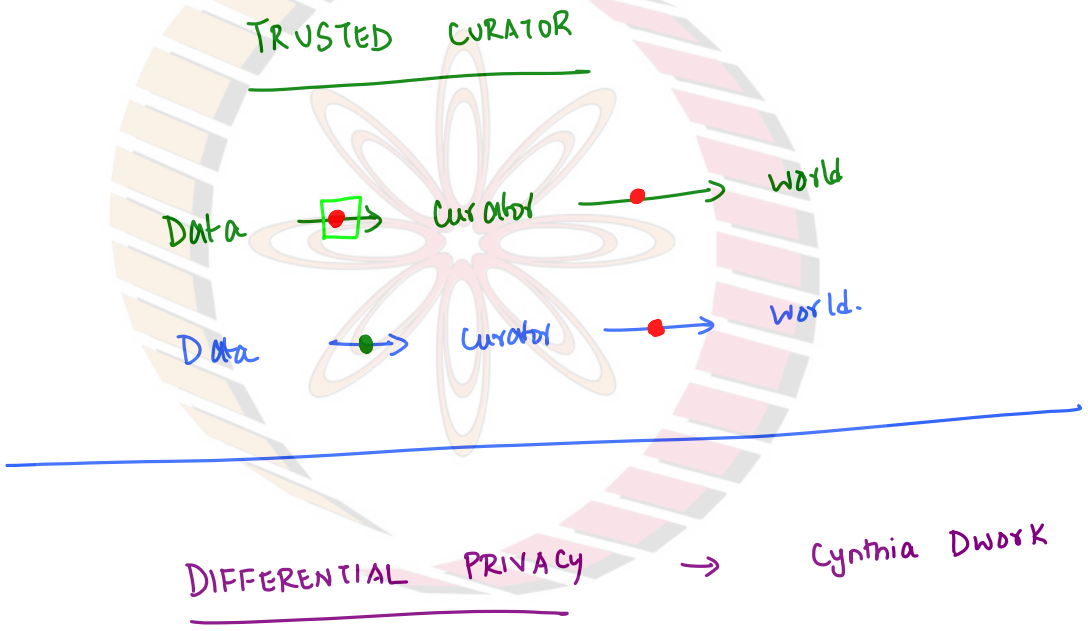
$\mathbb{E}[\bar{z}] = ? = \bar{x}$

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Note Title


Utility :
$\left| \frac{1}{n} \sum_{i=1}^{n} x_i - \frac{1}{n} \sum_{i=1}^{n} z_i \right|$
with high probability
$\leq O\left(\frac{1}{\epsilon \sqrt{n}}\right)$
UTILITY
ϵ Controls Privacy - utility tradeoff

Note Title



NPTEL

Note Title

$\{x_1, \cdots, x_n\}$ $\quad x_i \in \{0,1\}$

$RR(x) = \{y_1, \cdots, y_n\}$

$y_i = x_i$ w.p $\frac{e^\epsilon}{1+e^\epsilon}$

$= 1 - x_i$ $\quad \frac{1}{1+e^\epsilon}$

x, x'

Fix any $b \in \{0,1\}^n$

$\forall b \in \{0,1\}^n$ $\quad \frac{Pr(RR(x) = b)}{Pr(RR(x') = b)} \leq e^{\epsilon}$ ← Privacy

Utility

$x_1, \cdots, x_n$

$\downarrow \quad \downarrow$

$y_1, \cdots, y_n$

$\downarrow \quad \downarrow$

$z_1, \cdots, z_n$

$\hookrightarrow \frac{1}{n}\sum_{i=1}^{n} z_i$

w.h.p

$\left|\frac{1}{n}\Sigma x_i - \frac{1}{n}\Sigma z_i\right| \leq O\left(\frac{1}{\epsilon\sqrt{n}}\right)$

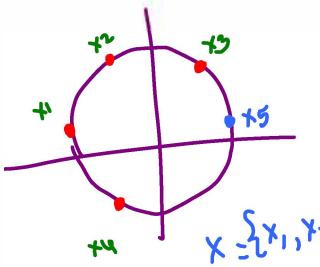
Note Title

Trusted Curator Model

Differential Privacy [Cynthia Dwork]

Let $M: X^n \rightarrow Y$. Consider two "neighboring" datasets x and $x' \in X^n$

$\{0,1\}^n \rightarrow \{0,1\}^n$



$X = \{x_1, x_2, x_3, x_4\}$

$x' = \{x_1, x_2, x_3, x_5\}$

M is ϵ- differentially private. if for all x, x' neighbouring.

and all $S \subseteq Y$

Note Title

$$\frac{Pr(M(x) \in S)}{Pr(M(x') \in S)} \leq e^{\epsilon}$$

In the previous example

$Y = \{0,1\}^n$ $\quad S \subseteq \{0,1\}^n$ $\quad S = \{b_1, b_{25}, b_{108}\}$

$$\frac{Pr(M(x) \subseteq S)}{Pr(M(x') \subseteq S)} = \frac{Pr(M(x) = b_1) + Pr(M(x) = b_{25}) + Pr(M(x) = b_{108})}{Pr(M(x') = b_1) + \cdots}$$

NPTEL

Note Title

LAPLACE MECHANISM

Sensitivity

$f : X^n \to \mathbb{R}$ (average)

$\Delta = \max_{x, x' \text{ neighbouring}} |f(x) - f(x')|$

x, x'

$$\left| \frac{1}{n} \sum_{i=1}^{n} x_i - \frac{1}{n} \sum_{i=1}^{n} x_i' \right|$$

$$\frac{1}{n} \left| \sum_{i=1}^{n-1} x_i + x_n - \sum_{i=1}^{n-1} x_i' - x_n' \right|$$

$$= \frac{1}{n} |x_n - x_n'|$$

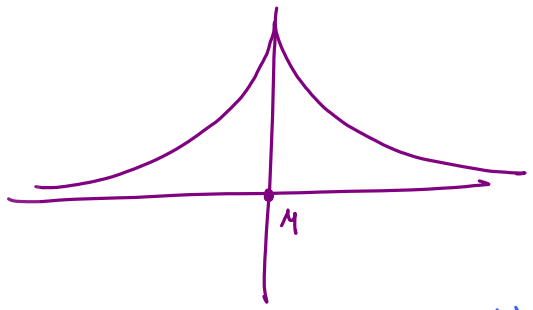
$$\Delta = \frac{1}{n}$$

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Note Title

LAPLACE DISTRIBUTION

$$f_{Lap}(x) = \frac{1}{2b} e^{-\frac{|x-\mu|}{b}}$$



Laplace mechanism : $f(x) + \eta$ (noise)

$\eta \sim Laplace\left(0, \frac{\Delta}{\epsilon}\right)$

Sensitivity of F

Privacy parameter

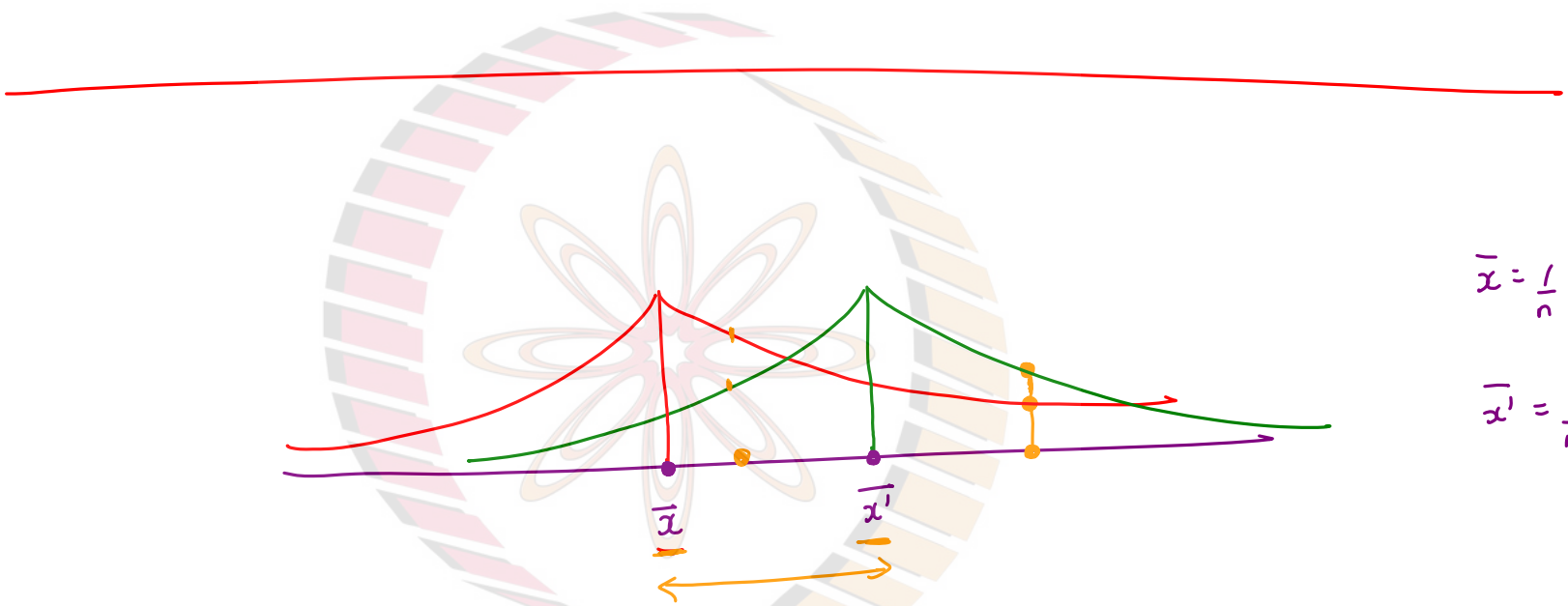
average + η

$\eta \sim Lap\left(0, \frac{1}{n\epsilon}\right)$

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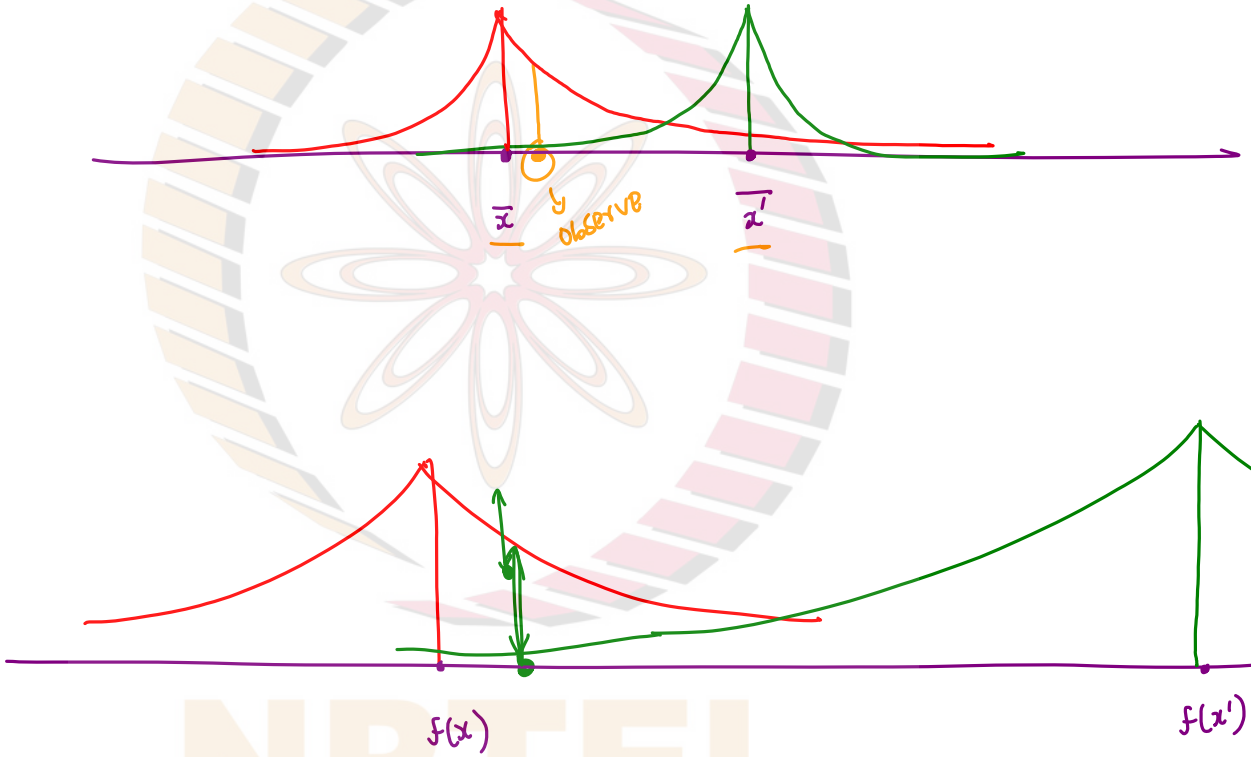
Note Title

- Can argue that the Laplace mechanism is ϵ-DP



$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

$$\bar{x'} = \frac{1}{n}\sum_{i=1}^{n} x'_i$$


x
y
OBSERVE
x'
f(x)
f(x')

Utility

Deviation from truth.

w.h.p $\left| \frac{1}{n} \sum_{i=1}^{n} x_i - \left(\frac{1}{n} \sum_{i=1}^{n} x_i + \eta \right) \right|$

Truth

actual released value $f(x) + \eta$

LAPLACE	R.R
$O\left(\frac{1}{\epsilon n}\right)$	$O\left(\frac{1}{\epsilon \sqrt{n}}\right)$

NPTEL